Chapter 1
The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Proofs

- Mathematical proofs are necessary in computer science for several reasons.
  - An program must always be proven correct.
  - You may also want to show that one program is more efficient than other program. This requires a proof.
  - Proving certain properties of data structures may lead to new, more efficient or simpler algorithms.

Proofs

The construction of a valid proof is an art, honed after much practice.

Some Terminology

- A proof is a valid argument that establishes the truth of a mathematical statement.
- A theorem (proposition/fact/result) is a statement that can be shown to be true.
- Axioms (postulates) are statements we assume to be true
- A lemma is a less important theorem that is helpful in the proof of other results
- A corollary is a theorem that can be established directly from a theorem that has been proved.
- A conjecture is a statement that is being proposed to be a true statement

  A conjecture becomes a theorem once it has been proved to be true.

Formal Proofs vs. Informal Proofs

- Formal proofs
  - Can be extremely long
  - May be hard to follow

- Informal Proofs
  - Each step may involve multiple rules of inference;
  - Steps may be skipped;
  - The axioms being assumed and the rules of inference used are not explicitly stated.

Informal proofs are designed for human consumption.

Understanding How Theorems Are Stated

1. Implied universal quantifiers

- Example

"if $x>y$, where x and y are positive real numbers, then $x^2>y^2$."

For all positive real numbers x and y, if $x>y$, then $x^2>y^2$.

2. Implicit implications

- Example

"The square of an odd integer is odd."

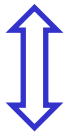
"if an integer n is odd, then n^2 is odd."

"For all integer n, if n is odd, then n^2 is odd."

Methods of Proving Theorems

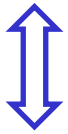
- Prove a theorem of the form

$$\forall x\,(P(x) \rightarrow Q(x))$$



Universal instantiation &
Universal generalization

$P(c) \rightarrow Q(c)$, where c is an arbitrary element of the domain



Truth table

When $P(c)$ is true, $Q(c)$ is true

Direct Proofs

$$P(c) \rightarrow Q(c)$$

- Assumes the hypotheses $P(c)$ are true
- Uses the rules of inference, axioms and any logical equivalences to establish the truth of the conclusion $Q(c)$

If n is odd, then n^2 is odd.

Proof:

Assume that n is odd.

Then $n = 2k + 1$, where k is some integer.

It follows that

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Therefore,

n^2 is odd (it is 1 more than twice an integer).

Q.E.D.

Proof by Contraposition

- A direct proof of the contrapositive

$$P \rightarrow Q \equiv \neg Q \rightarrow \neg P$$

  - Assumes that $\neg Q$ is true
  - Uses the rules of inference, axioms and any logical equivalences to establish $\neg P$ is true

A perfect number is not a prime.

A *perfect* number is one which is the sum of all its divisors except itself.
For example, 6 is perfect since 1 + 2 + 3 = 6.

Proof by contraposition:

We assume the number s is a prime and show it is not perfect.

But the only divisors of a prime are 1 and itself.

Hence the sum of the divisors less than s is 1 which is not equal to s.

Hence s cannot be perfect.

Q.E.D.

Vacuous Proof

- If we know P is false then $P \rightarrow Q$ is *vacuously* true.
- F $\rightarrow$ T and F $\rightarrow$ F are both true.
- Example:

  If Tom is both handsome and ugly then he feels unhappy.

Proof:

This is of the form $(p \wedge \neg p) \rightarrow q$

and the hypotheses $(p \wedge \neg p)$ form a contradiction.

Hence q follows from the hypotheses vacuously.

Q.E.D.

Trivial Proof

- If we know Q is true, then $P \rightarrow Q$ is true
- F $\rightarrow$ T and T $\rightarrow$ T are both true.
- Example:
  - If it's raining today then the empty set is a subset of every set.

  Proof:

  The assertion is *trivially* true independent of the truth value of ***p***.

Proof *p* by Contradiction

- Establish the truth of a statement *p* using the method of *proof by contradiction*
  - assumes *p* is false
  - deduces that $\neg p \rightarrow (q \wedge \neg q)$, which $q \wedge \neg q$ is a contradiction
  - *p* follows from the above

There is no largest prime number

Proof:

Let p be the proposition "there is no largest prime number."

Assume that $\neg p$ is true, namely, there is a largest prime number, denoted by s.

Hence, the set of all primes lie between 2 and s.

Form the product of these primes:

$$r = 2 \bullet 3 \bullet 5 \bullet 7 \bullet 11 \bullet .... \bullet s.$$

But $r + 1$ is a prime or has a prime factor larger than s.

This is a contradiction since we have shown that $\neg p$ implies both q and $\neg q$ where q is the statement that s is the largest prime number.

Hence, $\neg p$ is false, so that p, "there is no largest prime number," is true.

Q.E.D.

Proof $p \rightarrow q$ by Contradiction

- The proof of $p \rightarrow q$ by contradiction consists of the following steps:

1) assumes that both p and $\neg q$ are true

2) shows that $(p \wedge \neg q) \rightarrow$ F,

We have obtained a contradiction

Show that $s \vee r$ logically follows from the hypotheses $p \vee q, p \rightarrow r, q \rightarrow s$

Proof:

	Step	Reason
1.	$\neg (s \vee r)$	Additional hypothesis
2.	$\neg s \wedge \neg r$	De Morgan's Law, 1
3.	$\neg s$	Simplification, 2
4.	$\neg r$	Simplification, 2
5.	$p \rightarrow r$	Hypothesis
6.	$\neg p$	Modus tollens, 4, 5
7.	$q \rightarrow s$	Hypothesis
8.	$\neg q$	Modus tollens, 3,7
9.	$\neg p \wedge \neg q$	Conjunction, 6,8
10.	$\neg (p \vee q)$	De Morgan's Law, 9
11.	$p \vee q$	Hypothesis

contradiction

Q.E.D.

Proofs of Equivalence

(1) How to prove the proposition *"p* if and only if *q"*?

(2) How to prove that several propositions p_1 , p_2,...,p_n are equivalent?

To show that p_1 , p_2,...,p_n are equivalent,
establish the implications $p_1 \rightarrow p_2$, ..., $p_{n-1} \rightarrow p_n$, $p_n \rightarrow p_1$

$$[p_1 \leftrightarrow p_2 \leftrightarrow ... \leftrightarrow p_n] \equiv [(p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge ... \wedge (p_n \rightarrow p_1)]$$

Mistakes in Proofs

- Many mistakes result from the introduction of steps that do not logically follow from those that precede it.

- Many incorrect arguments are based on a fallacy called *begging the question* (circular reasoning).

Example: Prove that an integer n is even, if n^2 is even.

Attempted proof:

Assume n^2 is even. Then $n^2=2k$ for some integer k.

Let $n=2j$ for some integer j . This shows that n is even."

Begs the question: How can you show that $n=2j$ for some integer j, without first assuming n is even?

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Proof by Cases and Exhaustive Proof

- Proof by Cases
  - Break the premise of $p \rightarrow q$ into an equivalent disjunction of the form $p_1 \vee p_2 \vee \ldots \vee p_n$
  - Then use the rule

$$((p_1 \vee p_2 \vee \ldots \vee p_n) \rightarrow q) \equiv ((p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \ldots \wedge (p_n \rightarrow q))$$

Each of the implications $p_i \rightarrow q$ is a case.

- You must
  - Convince the reader that the cases are inclusive (i.e., they exhaust all possibilities)
  - Establish all implications

An *exhaustive proof* is a special type of proof by cases where each case involves checking a single example

Proof by Cases

- Examples:

1. If n is an integer not divisible by 3, then $n^2 \equiv 1 \pmod 3$.

Proof:

Let $P(n)$: n is an integer not divisible by 3; $Q(n)$: $n^2 \equiv 1 \pmod 3$

$P(n) \equiv (P_1(n) \vee P_2(n))$

where $P_1(n) : n \equiv 1 \pmod 3$; $P_2(n) : n \equiv 2 \pmod 3$

To show that $P(n) \rightarrow Q(n)$, we need to prove that both the cases $P_1(n) \rightarrow Q(n)$ and $P_2(n) \rightarrow Q(n)$ are true.

Case (i). $P_1(n) \rightarrow Q(n)$ (detail omitted)

Case (ii). $P_2(n) \rightarrow Q(n)$ (detail omitted)

Because $Q(n)$ holds in all two cases, we can conclude that If n is an integer not divisible by 3, then $n^2 \equiv 1 \pmod 3$.

Proof by Cases

- Examples:

2. Prove that if n is an integer not divisible by 2 or 3, then n^2-1 is divisible by 24.

   $n = 6k+1$

   $n = 6k+5$

3. Show that there are no solutions in integers x and y of $x^2+3y^2=8$.

   $x = -2,-1,0,1,2$

   $y = -1,0,1$

Existence Proof

- We wish to establish the truth of

$$\exists x P(x)$$

- *Constructive* existence proof:
  - Establish $P(c)$ is true for some c in the domain.
  - Then $\exists x P(x)$ is true by Existential Generalization (EG).

- *Nonconstructive* existence proof:
  - Assume no c exists which makes $P(c)$ true and derive a contradiction

Example (Constructive existence proof)

- Show that there are n consecutive composite positive integers for every positive integer n.

$$\forall n \exists x (x + i \text{ is composite for } i = 1, 2, \ldots, n).$$

Proof:

Let $x = (n + 1)! + 1$.

Consider the integers $x + 1, x + 2, \ldots, x + n$.

Note that $i + 1$ divides $x + i = (n + 1)! + (i + 1)$ for $i = 1, 2, \ldots, n$.

Hence, n consecutive composite positive integers have been given.

Q.E.D.

- Note that in the solution a number x such that $x + i$ is composite for $i = 1, 2, \ldots, n$ has been produced.

  Hence, this is an example of constructive existence proof.

Example (Nonconstructive existence proof)

- Prove that for all integers n, there exists a prime p so that $p > n$.

Proof:

Let n be an arbitrary integer, and consider $n! + 1$.

If $(n! + 1)$ is prime, we are done since $(n! + 1) > n$. But what if $(n! + 1)$ is composite?

If $(n! + 1)$ is composite then it has a prime factorization:

$\boldsymbol{p_1 p_2 \ldots p_k} = (n! + 1)$

Consider the smallest p_i, how small can it be?

Let $p = p_i$, then $p > n$, and we are done.

Q.E.D.

BUT WE DON'T KNOW WHAT p IS!!!

Uniqueness Proof

- To show that a theorem assert the existence of a unique element with a particular property.

$$\exists x\,(P(x) \wedge \forall\, y\,(y \neq x \rightarrow \neg P(y)))$$

There are two parts of a *uniqueness proof*:

- *Existence*: We show that an element *x with* the desired property exists.
- *Uniqueness* : We show that if $y \neq x$, then *y* does not have the desired property. Equivalently, we can also show that if *x* and *y* both have the desired property ,then $x=y$.

Uniqueness Proof

- Example

Show that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar + b = 0$.

Proof:

Existence: Note that the real number $r = -b/a$ is a solution of $ar + b = 0$ because $a(-b/a)+b=-b+b=0$. Thus, a real number r exists for which $ar + b = 0$.

Uniqueness: suppose that s is a real number such that $as + b = 0$. Then $as + b = ar + b$, where $r = -b/a$. Subtracting b from both sides, we find that $as = ar$. Dividing both sides of this last equation by a, which is nonzero, we see that $s = r$.

Q.E.D.

Proof Strategies

- Forward reasoning:

Using premises, together with axioms and known theorems to lead to the conclusion.

- Backward reasoning:

To reason backward to prove a statement q, we find a statement p that we can prove with the property that $p \rightarrow q$.

Example 1 (Backward Reasoning)

- Given two distinct positive real numbers x and y, their arithmetic mean is $(x+y)/2$ and their geometric mean is $\sqrt{xy}$, the arithmetic mean is always greater than the geometric mean.

Proof:

Suppose that x and y are distinct real numbers. Then

$$
\begin{aligned}
& (x-y)^2 > 0 \\
\equiv\ & x^2 - 2xy + y^2 > 0 \\
\equiv\ & x^2 + 2xy + y^2 > 4xy \\
\equiv\ & (x+y)^2 > 4xy \\
\equiv\ & (x+y)^2 / 4 > xy \\
\equiv\ & (x+y)/2 > \sqrt{xy}
\end{aligned}
$$

Q.E.D.

Example 2 (Backward Reasoning)

- Suppose that two people play a game taking turns removing, 1, 2, or 3 stones at a time from a pile that begins with 15 stones. The person who removes the last stone wins the game. Show that the first player can win the game no matter what the second player does.

Example 2 (Backward Reasoning)

Proof: Let *n* be the last step of the game.

Step n: $Player_1$ can win if the pile contains 1,2, or 3 stones.

Step n-1: $Player_2$ will have to leave such a pile if the pile that he/she is faced with has 4 stones.

Step n-2: $Player_1$ can leave 4 stones when there are 5,6, or 7 stones left at the beginning of his/her turn.

Step n-3: $Player_2$ must leave such a pile, if there are 8 stones.

Step n-4: $Player_1$ has to have a pile with 9,10, or 11 stones to ensure that there are 8 left.

Step n-5: $Player_2$ needs to be faced with 12 stones to be forced to leave 9,10, or 11.

Step n-6: $Player_1$ can leave 12 stones by removing 3 stones.

Now reasoning forward, the first player can ensure a win by removing 3 stones and leaving 12.

Methods of Proofs

- Direct proof
- Proof by contraposition
- Vacuous proof
- Trivial proof
- Proof by contradiction
- Proof by cases
- Existence proof
- Uniqueness proof
- Proof by counterexample

Additional Methods of Proofs

- Mathematical induction
- Structural induction
- The Cantor diagalization method
- Combinatorial proofs

Homework (Due on March 21)

Ed. 8

Sec. 1.7 7, 8, 36

Sec. 1.8 24, 26

Ed. 7

Sec. 1.7 7, 8, 34

Sec. 1.8 22, 24